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Trincomalee Campus, Eastern University, Sri Lanka
Faculty of Applied Science
Department of Computer Science
Bachelor of Computer Science (2017/2018)
Year I Semester II Examination (September, 2020)
CO 1225: Computer Architecture

Answer all questions **Time: 02 Hours**

Q1)

a) Convert the following hexadecimal numbers into decimal, binary and octal numbers.

i) BC79DE,

ii) 16CDA. **[2*12=24 Marks]**

b) Write the following decimal numbers in eight bits two's complement, do the addition/subtraction, and convert your answer back to decimal.

i) -85+11,

i) -72 + 15,

ii) -100-17. **[3*12=36 Marks]**

c) Represent the following numbers in Binary Coded Decimal code.

i) 357,

ii) 8579. **[2*6=12 Marks]**

d) What are the advantages of using Unicode over ASCII? **[12 Marks]**

e) Convert the following binary numbers into its decimal equivalent.

i) 0.1011_2,

ii) 1101.0111_2. **[2*8=16 Marks]**

Q2)

a) A logic circuit has four binary inputs D, C, B, A that represent the sixteen values 0 (0, 0, 0, 0) to 15 (1, 1, 1, 1). The output F is true if the decimal input on D, C, B, A is divisible by any of the number 3, 8, 13, or 14. Note that 0 is not divisible by any number. Draw a truth table for this system with inputs D, C, B, A and output F. **[40 Marks]**

b) From the truth table, write down an expression for the output F (unsimplified). **[20 Marks]**

c) Simplify the above equation obtained from part (b). **[20 Marks]**

d) Design a circuit using logic gates to implement the simplified expression for F. **[20 Marks]**

Q3)

a) How are computer manufacturers attempting to increase the performance of microprocessors? **[15 Marks]**

b) What is a cache memory and briefly explain how it improves performance? **[13 Marks]**

c) The main memory of a computer has an access time of 50 ns. The cache memory has an access time of 5 ns. The hit ratio for the memory system is 0.9 (i.e., 90% of memory accesses are to the cache). What is the speedup ratio of this system? **[15 Marks]**

d) Explain the function of the program counter. **[12 Marks]**

e) The operation of the simple Von Neumann computer with its fetch/execute cycle has been enhanced by several techniques to improve its performance. Write short notes on the following three mechanisms that are found in most computers. In each of the cases, explain the function using appropriate techniques.

   i) Virtual memory,

   ii) Direct memory access (DMA),

   iii) The interrupt. **[15*3=45 Marks]**

Q4)

a) Briefly describe the followings.

   i) ROM,

   ii) Instruction set architecture,

   iii) Pipelining,

   iv) Parallel processing,

   v) L1-Cache. **[5*08=40 Marks]**

b) Briefly explain the instruction execution cycle with the suitable diagram. **[15 Marks]**

c) Draw the diagram for Von-Neumann computer architecture. **[15 Marks]**

d) What is the stored-program concept and why is it important? **[15 Marks]**

e) Compare and contrast the "Complex Instruction Set Computers (CISC)" versus "Reduced Instruction Set Computer (RISC)". **[15 Marks]**